

Aim: Implement Linear Search to find an item in a list.

1.WAP to accept roll no of group of 20 student the teacher wants to search for the presence of a particular student in a group. Perform the linear search to find the student.

a. If student found return the position at which it is found (roll no start from 1)

b. If student not found then return 0.

c. Implement the search from the back of the array and provide a proper message/prompts.

CODE:

```
#include <iostream>
using namespace std;
int search(int n){
    int s[n];
    int r,i;
    cout<<"Enter the RollNo:";
    for(i=0;i<n;i++){
        cin>>s[i];
    }
    cout<<"Enter the RollNo you want to check:";
    cin>>r;
    for(i=n-1;i>=0;i--){
        if(s[i]==r){
            cout<<"RollNo is found at position "<<i+1<<".";
            return 0;
        }
    }
    cout<<"RollNo:"<<r<<" not found.";
    return 0;
}
int main(){
    int n,i,in;
    cout<<"Enter the number of student you want:";
    cin>>n;
    in = search(n);
    return 0;
}
```

OUTPUT:

```
Enter the number of student you want:20
Enter the RollNo:1 2 3 4 5 11 12 13 14 15 6 7 8 9 10 16 17 18 19 20
Enter the RollNo you want to check:14
RollNo:14 is found at position 8.
-----
Process exited after 37.11 seconds with return value 0
Press any key to continue . . . |
```

```
Enter the number of student you want:20
Enter the RollNo:1 2 3 4 5 11 12 13 14 15 6 7 8 9 10 16 17 18 19 20
Enter the RollNo you want to check:55

-----
Process exited after 31.73 seconds with return value 0
Press any key to continue . . . |
```

2.You are now given a list of country name as a member of United Nation your task is to find the name of a particular country from the list. WAP to perform the linear search for this task.

CODE:

```
#include <iostream>

#include <cstring>

using namespace std;

int searching(int n){

    char c[n][30];

    char find[30];

    int i;

    cout<<"Enter the country in UN:";

    for(i=0;i<n;i++){

        cin>>c[i];

    }

    cout<<"Enter the country:";

    cin>>find;

    for(i=0;i<n;i++){

        if(strcmp(c[i],find)==0){

            cout<<"Country:"<<find<<" is present.";

            return 0;

        }

    }

    cout<<"Country:"<<find<<" not found";
```

```
}  
  
int main(){  
  
    int n;  
  
    cout<<"Enter the number of country you want:";  
  
    cin>>n;  
  
    searching(n);  
  
    return 0;  
  
}
```

OUTPUT:

```
Enter the number of country you want:5  
Enter the country in UN:INDIA RUSSIA CHINA JAPAN GERMANY  
Enter the country:india  
Country:india is present.  
-----  
Process exited after 26.13 seconds with return value 0  
Press any key to continue . . . |
```

```
Enter the number of country you want:5  
Enter the country in UN:INDIA JAPAN CHINA RUSSIA GERMANY  
Enter the country:AMERICA  
Country:AMERICA not found  
-----  
Process exited after 23.91 seconds with return value 0  
Press any key to continue . . . |
```